THE SEPARATION AXIOMS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The base change formula for the diagonal. Invoked at the start of the counted proof (source lines 710-715) as a cross-chapter pointer. Identifies the diagonal of a base change with a base change of the diagonal.
- The stability under base change of the properties of the diagonal figuring in the definition of the separation axioms. Invoked at the end of the counted proof (source lines 716-723). Transfers each axiom from the morphism to its base change.

Source reference. Original reference categories-lemma-base-change-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 476–476. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-base-change-diagonal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original definition definition-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 601–601. Navigation pointer only: the counted text refers to the Stacks item definition-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the

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Lemma 0.1. *All of the separation axioms listed in Definition definition-separated are stable under base change.*

Proof. Let $f : \mathcal{X} \to \mathcal{Y}$ and $\mathcal{Y}' \to \mathcal{Y}$ be morphisms of algebraic stacks. Let $f' : \mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{Y}'$ be the base change of f by $\mathcal{Y}' \to \mathcal{Y}$. Then $\Delta_{f'}$ is the base change of Δ_f by the morphism $\mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}' \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$, see Categories, Lemma categories-lemma-base-change-diagonal. By the results of Properties of Stacks, Section stacks-properties-section-properties-morphisms each of the properties of the diagonal used in Definition definition-separated is stable under base change. Hence the lemma is true. □